From Furniture Photo to Scale Cut Sheet in 3 Steps

SCALES: 1:10 TO 1:50 + CUSTOM
NO 3D CAD REQUIRED

EN Page 1 — Workflow: choose the base shape → enter real cm and scale → print at 100 % and verify with the 5 cm ruler. Use the Neufert table for dimensions.

ع الصفحة ١ — سير العمل: اختر الشكل الأساسي ← أدخل الأبعاد الحقيقية والمقياس ← اطبع بنسبة ١٠٠٪ وتحقق بمسطرة ٥ سم. استخدم جدول نويفرت للأبعاد.

📐 **UNIVERSAL ARCHITECTURAL SCALE FORMULA (ANY SCALE 1 : X)**

The tool calculates this automatically, or you can check any dimension on your calculator:

Model Size (mm) = (Real Size in cm × 10) ÷ Scale

1 Pick Base Archetype

Look at your furniture reference photo. Match it to 1 of the **6 Core Shapes** in the tool (Chair, Sofa, Table, Bed, Cabinet, or Bookcase). Upload your photo to trace any curved backrest or arched headboard.

2 Enter Real CM & Scale

Set the real-life size in centimeters (use the Neufert chart below!). Choose **1:20** (where 1 m = 5 cm) or **1:10** (where 1 m = 10 cm). The tool converts everything to exact millimeters automatically.

3 Print 1:1 & Verify

Click **Print A4 Sheet** (set printer to *"100% / Actual Size"*). Place your metal studio ruler against the **5 cm Check Bar** in the corner of the page to verify your print scale is 100% accurate.

STANDARD INTERIOR DESIGN DIMENSIONS CHEAT SHEET (NEUFERT STANDARDS)

Furniture Piece	Real Width / Length	Real Depth	Real Height	Size on 1:20 Sheet	Size on 1:10 Sheet
Dining Chair	45 cm	45 cm	Seat 45 / Back 85 cm	22.5 × 22.5 × 42.5 mm	45 × 45 × 85 mm
3-Seater Sofa	200 cm	85 cm	Seat 42 / Back 80 cm	100 × 42.5 × 40 mm	200 × 85 × 80 mm
Dining / Study Table	140 – 180 cm	80 cm	75 cm	70 × 40 × 37.5 mm	140 × 80 × 75 mm
Queen Bed + Headboard	160 cm	200 cm	Base 35 / Head 110 cm	80 × 100 × 55 mm	160 × 200 × 110 mm
Credenza / Wardrobe	120 cm	50 – 60 cm	80 cm (or 210 cm)	60 × 25 × 40 mm	120 × 50 × 80 mm

RED = FULL CUT
BLUE = HALF SCORE

How to Cut, Score & Assemble Clean Maquettes

EN Page 2 — Craft: print on paper, mount on card, score the blue lines first, cut the red lines second, glue with a toothpick of PVA. Laser: use the layered SVG.

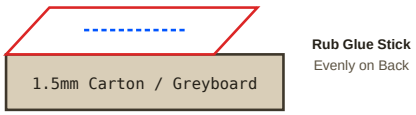
ع الصفحة ٢ — التنفيذ اليدوي: اطبع على ورق، ثبّت على كرتون، حزّز الخطوط الزرقاء أولًا، ثم اقطع الخطوط الحمراء، والصق بعود أسنان وغراء أبيض. الليزر: استخدم ملف SVG ذا الطبقات.

TECHNIQUE A — Normal Home/Office Printer

1. Print on Paper → Mount on Carton

EN Print on 80 g paper, spray-mount on 0.5–1.5 mm card. Never scale the print.

ع اطبع على ورق ٨٠ جم، والصقه برذاذ على كرتون ٠٫٥–١٫٥ مم. لا تغيّر مقياس الطباعة أبدًا.



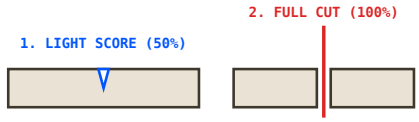
Option 1: Load **250g–300g Bristol card** directly into your normal paper printer.
Option 2: Print on normal 80g copy paper, apply a thin layer of **Glue Stick** (not wet liquid glue!) to the back, and press it flat onto **1.5mm Greyboard/Carton**.

GOLDEN RULE #1 — Crisp 90° Edges

2. Score BLUE Lines First, Cut RED Second!

EN Score the blue lines lightly with the back of the blade and a steel ruler, then cut red lines fully. Scoring first keeps folds crisp.

ع حزّز الخطوط الزرقاء بخفة بظهر الشفرة ومسطرة معدنية، ثم اقطع الخطوط الحمراء بالكامل. التحزيز أولًا يحافظ على الطيات حادة.



Always run your cutter lightly along all **Blue Dashed Lines FIRST** while the sheet is still one large, stable piece held under your steel ruler. Only cut the outer **Solid Red Lines** at the very end!

JOINERY — Zero Glue Stains

3. Toothpick PVA Glue & Slotted Joints

EN Apply a thin line of PVA with a toothpick on the tabs; hold 20 seconds. Slot joints need no glue.

ع ضع خطًا رفيعًا من الغراء الأبيض بعود أسنان على الألسنة؛ اضغط ٢٠ ثانية. الوصلات المحزّزة لا تحتاج غراء.

Never squeeze glue directly from the bottle onto your maquette! Put a drop of white wood/PVA glue on scrap card and apply tiny dots to the grey glue tabs using a **wooden toothpick**. Hold corners square for 15 seconds.

LASER / CNC — Adobe Illustrator Ready

4. Using the Downloaded SVG on a Laser Cutter

EN Open the SVG in Illustrator / laser software. Layer 01 = cut, 02 = score (low power), 03 = labels (engrave or delete).

ع افتح ملف SVG في إليستريتور أو برنامج الليزر. الطبقة 01 = قص، 02 = تحزيز (طاقة منخفضة)، 03 = كتابات (نقش أو حذف).

Click **"Download SVG / XML Vector"** and open the file in **Adobe Illustrator**. The file is already pre-grouped into 3 standard laser layers: `01_CUT_RED`, `02_SCORE_BLUE`, and `03_LABELS_BLACK`.

Make the Image the Tool Can Read

The tracer in **Tab 3** measures a flat picture. It can only be as accurate as the image you give it. The ideal input looks like an **architect's elevation drawing**: one view, straight-on, no perspective, plain white background, no shadow, no floor, the whole object visible, in a solid dark colour. Below are two ready prompts — one to **generate** such an image, one to **convert** an image you already have.

✅ Tool-ready image checklist

- **Side or front elevation**, camera at mid-height, looking straight at the piece (orthographic). Verticals must be vertical.
- **Plain, even background** — pure white is best. Any single flat colour also works.
- **No cast shadow, no floor line, no reflection, no room.**
- **Whole object inside the frame** with a margin; nothing cropped.
- **One piece per image.** Dark solid colour, no texture or pattern needed.
- Legs, arms and back must be **visibly separated** from the background (a dark chair on white, never white on white).

❌ What makes the trace wrong

- **¾ perspective / "product shot" angle** — the far side is foreshortened; your scale will be wrong even after calibration.
- **Room scenes** (rug, wall, window) — the background cannot be separated. Use *Hand-trace* or convert the image first.
- **Soft drop shadow glued to the legs** — the tool removes most of it; a very dark shadow still needs the "Clean edges" slider or a point deleted by hand (Alt-click).
- **Transparent chairs, glass, chrome, white-on-white** — no contrast, no outline.
- **Cushions, throws, plants** on the piece — they become part of the silhouette.

Prompt A — GENERATE a new tool-ready image (Gemini, ChatGPT, Copilot, Ideogram…)

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Create a technical 2D SIDE ELEVATION drawing of a [DINING CHAIR / LOUNGE SOFA / COFFEE TABLE / BED / WARDROBE] in a [Scandinavian / Mid-century / Minimalist] style.

Strict requirements:
- View: exact side elevation, orthographic projection, camera at mid-height, no perspective, no 3/4 angle, no foreshortening. All vertical edges perfectly vertical.
- Background: pure flat white (#FFFFFF). No gradient, no vignette, no room, no wall, no floor line.
- No shadow, no reflection, no ground contact shadow, no ambient occlusion.
- Whole object fully inside the frame with 10% white margin on every side, centred, nothing cropped.
- Render the object in one solid dark matte colour (charcoal #2B2B2B) with crisp clean edges. No texture, no wood grain, no fabric pattern, no highlights.
- Nothing else in the image: no text, no labels, no dimension lines, no watermark, no people, no cushions, plants or accessories.
- Realistic proportions: seat height 45 cm, total height 85 cm (adjust to the piece).
- Square image, 2048 × 2048 px.

Then produce a second image with the same object and the same rules as a FRONT ELEVATION.
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Edit the words in [brackets]. Ask for the **side** view for chairs, sofas and beds (it gives the profile you cut twice), and the **front** view for shelves, cabinets and tables.

Prompt B — CONVERT an image you already have (upload it to Gemini, then paste this)

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Use the attached image as the reference. Keep the exact same piece of furniture — same design, same proportions, same number of legs, same back and arm shape — but redraw it as a clean technical 2D SIDE ELEVATION.

Strict requirements:
- Remove the entire background, room, floor, wall, rug and all other objects. Replace with pure flat white (#FFFFFF).
- Remove every shadow and reflection, including the ground-contact shadow under the legs.
- Rotate the view to an exact side elevation (orthographic, no perspective, camera at mid-height). Do not change the furniture's real proportions while doing this: a 45 cm seat stays at 45 cm relative to an 85 cm back.
- Draw the object as one solid dark matte charcoal silhouette (#2B2B2B) with crisp edges. No texture, no highlights, no colour, no outline strokes.
- Remove cushions, blankets, plants, books and any accessories that are not part of the furniture structure.
- Whole object fully visible and centred with a 10% white margin; nothing cropped.
- No text, labels, dimension lines or watermark. Square image, 2048 × 2048 px.

If the reference shows a 3/4 view, infer the true side elevation from it; do not keep the angled view.
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Check the result against the ✅ checklist before you upload it to Tab 3. If Gemini keeps the shadow, reply: *"Remove the shadow under the legs completely; the background must be 100% white everywhere."*

1 · Upload
Drop the AI image into Tab 3 (or Ctrl+V).

2 · Calibrate
Click floor → seat top (45 cm), or floor → top (85 cm). The image has no real size until you do this.

3 · Auto-trace
Method "Remove background". Not clean? raise tolerance, or 💧 pick the background, or Hand-trace.

4 · Sheet
Scale 1:10 / 1:20, 2 copies for sides, add seat / top panels in cm, Print 100% or Laser SVG.

Academic honesty note: the AI image is only a *drawing aid*. Your dimensions must come from the Neufert table on page 1 (or a measured piece), and you must state in your submission which AI tool and prompt produced the reference image.